INNOVATHON 2026

Challenge 1 of 5

Zero-Touch Growth Operating System

Track	Web, App & Platform Development
Difficulty Level	Medium
Team Size	3–4 members
Duration	24 hours
Hackathon Date	April 1, 2026 – April 2, 2026
Prepared for	NMIMS INNOVATHON 2026

IMPORTANT NOTICES & POLICIES

- API Keys Security: Never commit API keys to GitHub. Use .env.local for local development. Vercel deployment: set keys in Settings → Environment Variables.
- Original Work: All code must be written or substantially modified during the hackathon. Pre-built, fully complete solutions are disqualified.
- AI Coding Tools: Use of GitHub Copilot, Cursor, Claude, or ChatGPT for code generation is permitted. Final solution must be understood and explainable by team members.
- Gemini API Fallback: If Gemini API is unavailable during hackathon, use cached sample response JSON or Claude free tier. Provide fallback implementation.
- Team Registration: Register your team at NMIMS Hackathon Portal before March 19, 2026, 5 PM IST.
- Disqualification Criteria: Using paid APIs, submitting pre-built solutions, or incomplete registration will result in disqualification.

1. EXECUTIVE SUMMARY

Build an autonomous digital growth platform that replaces the need for a full-time social media manager or external digital marketing agency. The system should intelligently understand a business, generate natural-looking creative assets, automate social media management, plan weekly campaigns based on market trends and festivals, and optimize ad performance end-to-end with minimal human intervention.

This is not another scheduling tool. This is a growth operating system that thinks like a strong marketing team.

2. THE PROBLEM STATEMENT

2.1 Core Challenge

- Small businesses and startups cannot afford full-time social media managers (₹30,000–50,000/month) or external agencies (₹50,000–2,00,000/month).
- Existing tools (Buffer, Later, Hootsuite) only handle scheduling and analytics — they do not create intelligent campaigns or optimize ads in real time.
- Business owners waste 3–5 hours per week manually creating content, uploading posts, monitoring performance, and making decisions.
- Marketing campaigns are not aligned with market dynamics, festivals, occasions, or emerging trends — missing critical revenue opportunities.
- No platform exists that combines social media operations, creative generation, market intelligence, and ad optimization into one autonomous workflow.

2.2 Market Opportunity

- ₹2,500+ crore Indian digital marketing market growing 28% annually.
- 50 million+ SMEs in India with zero organized digital marketing infrastructure.
- Existing AI social tools (Jasper, Copy.ai, Buffer AI) already generate captions, proving market validation.
- No single platform combines all into one end-to-end autonomous system.

3. WHAT TO BUILD — Core Modules

3.1 Business Intelligence Layer

The system ingests and analyzes:

- Business website URL and landing pages, industry category and business model
- Target audience persona and buying intent, competitor positioning
- Seasonal demand patterns and festival calendar, current trending topics and market dynamics

Output: AI generates a structured business profile and market analysis within 2 minutes.

3.2 Weekly Growth Planner

System creates an intelligent weekly execution plan considering:

- Business goals for the week, product launches or special announcements
- Upcoming festivals and occasions (Diwali, Valentine's Day, etc.)
- Trending topics in the industry, competitor activity and market gaps
- Audience fatigue from previous campaigns and past performance data

Output: AI generates a 7-day content calendar with themes, post types, timings, and campaign focus.

3.3 Social Media Automation Module

- Generate captions in brand tone with hashtags and CTAs
- Suggest optimal posting times based on audience activity
- Schedule posts across platforms (Instagram, LinkedIn, Facebook, Twitter)
- Auto-respond to common comments with branded templates
- Track engagement metrics (likes, comments, shares, reach)
- Re-queue evergreen content when needed
- Provide daily engagement summary to business owner

Output: Posts scheduled, published, and monitored autonomously. Business owner receives daily digest via email or WhatsApp.

3.4 AI Creative Generation Module

- Post captions (3–5 variations for A/B testing)
- Static post images (fetched from Unsplash, customized with text overlays)
- Festival creatives (pre-designed templates with dynamic text)
- Carousels (3–5 slide educational or product posts)
- Ad banners (multiple sizes for different platforms)

Output: Studio-quality creatives that look professionally designed, not obviously AI-generated.

3.5 Ad Performance Management Module

- Create ad copy variations and targeting audiences automatically
- Launch campaigns with objective-based settings (leads, conversions, awareness)
- Monitor KPIs in real time (CTR, CPC, CPL, ROAS, conversion rate)
- Identify winning audiences, creatives, and messaging
- Recommend budget reallocation to high performers
- Pause underperforming campaigns automatically

Output: Ad dashboard showing performance metrics, optimization recommendations, and projected outcomes.

4. MINIMUM VIABLE PRODUCT (MVP) — 24-Hour Build

Teams should deliver a working prototype covering these core features:

Feature	What to Deliver
Business Onboarding Form	Input: website URL, business type, target audience, location, products/services. Output: AI-generated business profile.
Weekly Content Calendar	AI generates 7-day calendar with post themes, timing, content pillars based on business + upcoming festivals.

AI Caption Generator	Paste a product description → system generates 3 caption variations + hashtags + CTA in brand tone.
Festival/Trend Detector	System scans upcoming festivals (30 days) and suggests campaign ideas aligned with business.
Ad Recommendation Engine	Mock ad dashboard (simulated data). Show Ad recommendations based on audience, budget, goal with AI insights.
Performance Dashboard	Mockup showing sample metrics: CTR, CPC, ROAS, conversion rate with AI-generated insights.
Integration Demo	End-to-end workflow: Business signup → calendar generated → captions created → recommendations shown.

Video generation and live Meta/Google Ads API integration are stretch goals, NOT required for MVP.

5. RECOMMENDED TECH STACK

All tools below are FREE with no infrastructure cost. Total Cost: ₹0

Component	Tool	Why This
AI Engine	Google Gemini 1.5 Flash (Google AI Studio)	1M tokens/day free, 15 req/min, fast response
Image Sourcing	Unsplash API	50 req/hour free, unlimited library
Frontend	Next.js 14 + React + TypeScript	Free, deployed on Vercel free tier
Backend	Next.js API routes (serverless)	Free, no separate server
Database	Firebase Firestore	50K reads/day free tier
File Storage	Firebase Storage	5 GB free
Deployment	Vercel	Free tier, auto-deploy from GitHub
UI Components	Tailwind CSS + shadcn/ui	Free, open source

5.1 API Rate Limiting & Fallback

- Cache responses: Store generated captions, calendars in browser localStorage for 1 hour
- Fallback JSON: If API fails, show pre-seeded sample response (stored in code)
- User feedback: Show error message: 'Rate limit reached. Using cached suggestion.' with timestamp
- Graceful degradation: Show loading state, don't break UI

6. PRE-REQUISITES FOR STUDENTS

6.1 Knowledge Requirements

Teams should have experience with:

- Frontend: React, TypeScript, CSS (Tailwind)
- Backend: API routes, database queries
- API integration: REST APIs, JSON, error handling
- Prompt engineering: Writing effective AI prompts
- UX/UI basics: Clean, intuitive interfaces

6.2 Tools Setup (30 Minutes)

- Google Account: Any Gmail account works
- Node.js & npm: nodejs.org, verify with npm -v
- Git & GitHub: github.com account created
- Code Editor: VS Code (code.visualstudio.com)
- Gemini API Key: aistudio.google.com → Get API Key (2 minutes, no card required)
- Unsplash API Key: unsplash.com/developers → Create free app (2 minutes)
- Firebase Project: firebase.google.com → New project → Copy config (5 minutes)

7. 24-HOUR BUILD TIMELINE

Hours 0–2: Setup & Architecture

- Clone starter boilerplate, npm install && npm run dev
- Create .env.local with API keys (use .env.example as template)
- Test all API connections. Assign team roles: Frontend, Backend, AI/Prompt Engineer, UI/UX

Hours 2–10: Core Features (Business Profile → Calendar → Captions)

- Business Onboarding form: 2 hours (input validation, Gemini integration)
- Weekly Planner: 3 hours (Gemini generates calendar, festival awareness)
- Caption Generator: 2 hours (3 variations, brand tone)
- UI Dashboard: 1 hour. GitHub commits every 2 hours.

Hours 10–18: Secondary Features & Testing

- Festival/Trend detector: 2 hours
- Mock Ad Recommendations: 2 hours (simulated data, no real API)

- Performance Dashboard: 1 hour (charts with Plotly or Recharts)
- Error handling & UI polish: 2 hours. Test full workflow on real sample business: 1 hour.

Hours 18–24: Testing, Deployment & Demo

- Final testing: 1 hour (all features working)
- Deploy to Vercel: 30 minutes (GitHub push auto-deploys)
- Create README with setup & usage: 1 hour
- Prepare demo: 3 sample businesses with full workflow: 1 hour
- Presentation prep: 1.5 hours (problem, solution, demo, roadmap). Final commits: 30 minutes.

8. SUBMISSION CHECKLIST

- Working web app deployed to Vercel (live link in README)
- GitHub repository with clean commit history
- README.md with: setup instructions, API key setup, usage examples, demo business data
- Functional MVP covering at least 5 of 7 core features
- Live demo showing end-to-end workflow on 3 sample businesses
- 5-minute presentation: Problem → Solution → Demo → Business Model → Roadmap
- Code is readable, commented, follows best practices. No API keys in source code.

9. JUDGING CRITERIA & SCORING

Criteria	Weight	What Judges Evaluate
Innovation & Uniqueness	20%	Does the solution replace manual marketing work? Does it stand out from existing tools?
Technical Execution	25%	Code quality, API integration, error handling, scalability, deployment, reproducibility.
User Experience	20%	Interface intuitive? Outputs look professional? Demo is smooth?
Business Intelligence	15%	Does system generate relevant, India-specific, festival/trend-aware content? Real business value?
Completeness	10%	How many features implemented? Is MVP feature-complete? Does everything work?
Presentation & Communication	10%	Clear problem statement? Good demo? Can team

		explain architecture and limitations?

10. FREQUENTLY ASKED QUESTIONS

Q1: Do we need Meta or Google Ads API?

No. Use simulated/mock ad data for MVP. Show recommendations only. Real API integration is a stretch goal.

Q2: Can we use Claude or Perplexity instead of Gemini?

Yes. Just ensure 1M tokens/day free quota. Gemini is optimal. Document your choice in README.

Q3: What if Gemini API goes down during hackathon?

Fallback: Use cached response JSON (pre-seeded in code) or switch to Claude free tier. Be prepared!

Q4: Do we need video generation?

No. Skip videos for MVP. Show 'Coming Soon' placeholder. Video generation is a stretch goal.

Q5: Can we use a different tech stack?

Yes, as long as ALL tools remain FREE. If any paid service required, it's disqualified.

Q6: How do we handle API keys securely?

Use .env.local for dev. On Vercel, use Settings → Environment Variables. Never commit keys to Git.

KEY TAKEAWAYS

- Problem: Businesses can't afford full-time marketers or agencies
- Solution: AI-powered autonomous growth operating system
- Tech: 100% free (Gemini, Firebase, Vercel, Unsplash)
- Track: Web, App & Platform Development
- Timeline: 24 hours for MVP | Scope: 7 features, min 5 required
- Judging: 6 factors, 100 points total | Cost: ₹0 students, ₹0 Vibe6 infra
- Key Success Factor: Good prompt engineering + real Indian business examples

Support: innovathon@nmims.edu | Register: NMIMS Portal | Registration Deadline: March 19, 5 PM IST

VIBE6

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INNOVATHON 2026

Challenge 2 of 5

Self-Hosted LLM Crypto Sentiment & Price Prediction Terminal

Track	Artificial Intelligence & Data Intelligence
Difficulty Level	HIGH (ML + Data Engineering)
Team Size	3–4 members
Duration	24 hours
Hackathon Date	April 1, 2026 – April 2, 2026
Prepared for	NMIMS INNOVATHON 2026

IMPORTANT NOTICES & POLICIES

- Educational Use Only: This is a simulation/educational tool. NO real money trading.
- API Keys: Never commit API keys to GitHub. Use .env.local (dev) and Environment Variables (production).
- Original Work: All code written during hackathon. Pre-built solutions disqualified.
- AI Tools: GitHub Copilot, Claude, ChatGPT permitted for code generation. Final solution must be understood and explained by team.
- Python Scripts: Submit as runnable Python scripts or Dockerized app. Jupyter notebooks alone NOT accepted.
- Disqualification: Using paid APIs, pre-built complete solutions, or incomplete registration = disqualified.
- Team Registration: Register at NMIMS Hackathon Portal by March 19, 2026, 5 PM IST.

1. EXECUTIVE SUMMARY

Build a self-hosted crypto trading intelligence terminal using open-source LLMs to analyze social media sentiment, news feeds, on-chain data, and historical price patterns to predict short-term crypto price movements. The system should run entirely locally (no expensive cloud APIs), provide real-time sentiment scoring, generate actionable trading signals, and help traders make data-driven decisions.

This is a full-stack ML + data engineering + trading systems project combining NLP, time-series forecasting, and real-time data pipelines.

2. THE PROBLEM STATEMENT

2.1 Core Challenge

- Crypto prices are highly influenced by social sentiment, news, and market psychology — not just technical factors.
- Retail traders have no automated way to aggregate sentiment from Reddit, Discord, and news feeds in real time.
- Professional sentiment APIs (Santiment ₹5L/month, LumiBot ₹2L/month) are inaccessible to retail traders.
- Existing crypto bots optimize technical indicators only — they ignore sentiment and narrative analysis.
- No open-source, self-hosted solution combines LLM sentiment analysis with price prediction.
- Traders waste hours manually analyzing Reddit posts and news articles instead of having AI do it automatically.

2.2 Market Opportunity

- Global crypto trading volume: $100+ billion daily. Indian crypto traders: 50 million+.
- Sentiment APIs in crypto: Santiment (₹5L/mo), LumiBot (₹2L/mo), CryptoQuant (₹1.5L/mo).
- Self-hosted LLM trend: Ollama, Hugging Face, LocalAI make running LLMs locally practical and free.
- Open-source LLMs (Mistral, Llama 2, Phi) now capable enough for financial NLP.

3. WHAT TO BUILD — Core Components

3.1 Data Ingestion Pipeline

Real-time collection from multiple sources:

- Reddit API: Stream posts and comments from r/cryptocurrency and coin-specific subreddits
- News API: Aggregate crypto news from CoinDesk, Cointelegraph, The Block, Crypto Briefing
- On-chain data: Whale transactions, exchange flows (Etherscan, Solscan free APIs — 5 calls/sec limit, 100K/day)
- Price data: Historical OHLCV from Binance free API (15-minute candles)

3.2 Sentiment Analysis Module (Self-Hosted LLM)

- Run Ollama locally: Mistral 7B or Llama 2 7B (requires 8GB VRAM GPU or 16GB RAM CPU)
- Classify Reddit posts/news as: BULLISH / NEUTRAL / BEARISH / FUD (sentiment score 0–1)
- Few-shot prompting: Use examples in prompt for better accuracy (~70–75% on finance text)
- LLM Fallback: If Ollama fails, use pre-trained FinBERT (Hugging Face, CPU-friendly)

3.3 On-Chain Analytics Module

- Whale transactions: Alert when large holders move >$1M
- Exchange inflows/outflows: Accumulation (outflow) vs. distribution (inflow) signals
- Combine with sentiment: 'Whales accumulating + sentiment bullish = strong buy'

3.4 Price Prediction Module (Time-Series ML)

- Input features: historical prices (15-min candles), volume, sentiment, on-chain metrics
- Models: Prophet (Facebook, CPU-friendly) OR LSTM (TensorFlow, GPU-friendly) OR XGBoost
- Prediction: Next 1h, 4h, 24h direction (UP/DOWN/SIDEWAYS) with confidence % (target >55%)
- Backtest on 6 months historical data: Track accuracy, precision, recall, Sharpe ratio

3.5 Trading Signal Generator

- BUY signal: Sentiment bullish (>0.7) AND prediction UP AND whales accumulating
- SELL signal: Sentiment bearish (<0.3) AND prediction DOWN AND whales withdrawing
- Explainability: Show user WHY each signal was generated (confidence breakdown)

3.6 Real-Time Dashboard/Terminal UI

- Multiple crypto tickers: BTC, ETH, SOL, major alts (at least 5)
- Real-time sentiment heatmap: color-coded sentiment per coin
- Signal log: chronological BUY/SELL with reasoning + outcomes
- Alert system: console/desktop notifications for high-confidence signals

4. MINIMUM VIABLE PRODUCT (MVP) — 24-Hour Build

Teams must deliver working prototype covering at least 5 of 8 core features:

Feature	What to Deliver
Data Ingestion	Real-time pipeline from 2+ sources (Reddit + Binance API). Store in PostgreSQL.
Sentiment Analysis	Run Mistral 7B or Llama 2 locally (or FinBERT as fallback). Classify 50+ samples.
Price Model	Train time-series model on 3 months historical data. >55% directional accuracy = good.
On-Chain Data	Fetch whale transactions from Etherscan free API. Display top recent transactions.
Signal Generator	Combine sentiment + prediction to generate BUY/SELL signals with reasoning.
Backtesting	Show hypothetical P&L of signals on past 30 days. Calculate win rate, Sharpe ratio.
Dashboard/Terminal	CLI (Rich library) or web (Streamlit) showing live data for

	3–5 coins.
Documentation	README with setup instructions, architecture diagram, API limits, model performance.

5. RECOMMENDED TECH STACK

All tools are FREE and open-source. No paid dependencies. Total Cost: ₹0

Component	Tool / Library	Notes
LLM (Self-Hosted)	Ollama + Mistral 7B or Llama 2 7B	GPU: 8GB VRAM. CPU: 16GB RAM. Fallback: FinBERT
Data Collection	PRAW (Reddit), NewsAPI (free tier), web3.py	All free tier, rate-limited but sufficient
Data Storage	PostgreSQL (Docker) or SQLite	Free, fast time-series queries
Time-Series Model	Prophet (Facebook) — CPU friendly	Train: 30 sec. Excellent for finance.
Alternative Model	LSTM (TensorFlow) — GPU recommended	Train: 5 min (GPU) / 30 min (CPU)
Dashboard	Rich (Python CLI) OR Streamlit (web)	Rich: beautiful terminal UI. Streamlit: quick web app.

6. PRE-REQUISITES FOR STUDENTS

6.1 Knowledge Requirements

This is a HIGH difficulty challenge. Teams should have:

- Python 3.9+: Strong proficiency (async, data structures, libraries)
- Machine Learning: LSTM, time-series forecasting, model evaluation, overfitting
- NLP: Text classification, sentiment analysis, prompt engineering
- APIs: REST API integration, rate limiting, error handling, authentication
- Data Engineering: Pipelines, streaming, ETL, data quality

6.2 Hardware Requirements

- Minimum 8GB RAM (16GB preferred) — for Mistral 7B locally
- 4GB+ free disk space — for LLM model weights (~7GB)

- GPU optional but HIGHLY recommended — M1/M2 Mac, NVIDIA RTX 3060+
- Stable internet — real-time Reddit, news, blockchain data streams

7. JUDGING CRITERIA & SCORING

Criteria	Weight	What Judges Evaluate
Technical Complexity	**30%**	Data pipeline quality, LLM setup, model sophistication, architecture design
Accuracy & Performance	**25%**	Model accuracy (>55% = good), backtested Sharpe ratio, signal win rate
Integration & Completeness	**20%**	How well do sentiment, prediction, on-chain data integrate? MVP feature-complete?
Code Quality & Documentation	**15%**	Readable code, error handling, documentation clarity, reproducibility
Presentation & Explanation	**10%**	Can team clearly explain the approach? Is live demo impressive?

KEY TAKEAWAYS

- Problem: Retail traders need affordable sentiment + forecasting tools
- Solution: Self-hosted LLM terminal combining sentiment, price forecasting, on-chain data
- Tech: 100% free (Ollama, Prophet/LSTM, PRAW, Binance API, PostgreSQL)
- Track: Artificial Intelligence & Data Intelligence
- Timeline: 24 hours for MVP with experienced ML teams
- Judging: 5 factors (Tech 30%, Accuracy 25%, Integration 20%, Code 15%, Presentation 10%)
- Cost: ₹0 students | Difficulty: HIGH — requires strong ML + data engineering + trading knowledge

Support: innovathon@nmims.edu | Register: NMIMS Portal | Registration Deadline: March 19, 5 PM IST

INNOVATHON 2026

Challenge 3 of 5

Micro Equity Tokenisation Platform for Indian MSMEs

Track	Blockchain, Fintech & Emerging Technologies
Difficulty Level	HIGH (Blockchain + Smart Contracts + Fintech Regulation)
Team Size	3–4 members
Duration	24 hours
Hackathon Date	April 1, 2026 – April 2, 2026
Context	Educational Prototype - Testnet Only - No Real Money

HACKATHON & EDUCATIONAL PURPOSE ONLY — This is a PROTOTYPE/SIMULATION. NOT a working securities offering, investment product, or financial service.

CRITICAL LEGAL DISCLAIMERS

1. Regulatory Disclaimer — SEBI & India Jurisdiction

- India Securities Regulation: This platform simulates equity tokenisation, which is REGULATED by SEBI.
- NOT a Real Securities Offering: PROTOTYPE ONLY. NOT a real securities offering under Indian law.
- SEBI Alternative Investment Fund (AIF) Regulations: Real equity crowdfunding requires SEBI AIF registration.
- Prevention of Money Laundering Act (PMLA) 2002: Real platforms must implement full KYC/AML/CFT. This prototype uses MOCK KYC only.
- Jurisdiction: Real operations require SEBI registration + Securities Contracts Act + Depositories Act + PMLA + Income Tax Act.

2. Blockchain & Technical Disclaimer

- Testnet Only: This platform operates EXCLUSIVELY on Polygon Mumbai testnet. Any mainnet deployment is STRICTLY PROHIBITED and results in IMMEDIATE DISQUALIFICATION.
- Test MATIC: All transactions use 'test MATIC' from faucet. NO real value.
- Smart Contract Status: Educational demonstrations ONLY. NOT for production use.

1. EXECUTIVE SUMMARY

Build a blockchain-based micro equity tokenisation platform enabling Indian MSMEs to raise capital by issuing fractional ERC-20 equity tokens to micro-investors (₹100 minimum). All transactions managed on-chain via smart contracts. This is NOT a real securities offering — it's a PROTOTYPE demonstrating technology feasibility for India's 63 million MSMEs who have no access to angel/VC funding or stock markets.

2. THE PROBLEM STATEMENT

2.1 Core Challenge

- 63 million Indian MSMEs contribute 30% GDP but 80% have NO access to formal equity capital.
- Bank loans require collateral. VC requires high growth. Angel investors inaccessible for small-town businesses.
- Traditional IPO minimum: ₹10 crore+ — completely out of reach.
- Investors cannot invest ₹100–₹10,000 in local businesses they understand.
- NO transparent, tamper-proof record of MSME equity ownership in India.
- NO secondary market for MSME equity — no exit for investors.

2.2 India-Specific Context

- SEBI regulatory sandbox allows fintech innovation.
- MSME sector: ₹43 lakh crore annual contribution to Indian economy.
- 500 million smartphone users — digital-first retail investors.
- Polygon blockchain: Near-zero gas fees ideal for micro-transactions.

3. WHAT TO BUILD — Core Modules

3.1 MSME Business Listing & Onboarding

- Business profile: Name, category, city, GST number, founding year
- Fundraising terms: Target amount (₹1L–₹50L), equity % offered (1–20%)
- IPFS storage: Decentralised, tamper-proof document access

3.2 Smart Contract Engine

- Token issuance: Unique ERC-20 per MSME — 1 token = fractional equity stake
- Investor KYC: Whitelist approved wallets (mock for MVP)
- Dividend distribution: Auto-distribute profits to token holders
- Governance: Token-weighted voting on business decisions
- Vesting: Founder token lock-up (6–12 month)
- Refund mechanism: Auto-refund if target not met

3.3 Investor Dashboard

- Browse MSME listings: Filter by city, sector, risk level
- Token purchase: Buy fractional equity (₹100+ minimum)
- Portfolio tracker: Holdings, valuation, dividend history

- Governance dashboard: View votes, cast vote

3.4 Risk Scoring

- AI-powered risk assessment based on revenue, age, sector, location
- Risk labels: Conservative / Moderate / High / Speculative

4. MVP SCOPE — 24-Hour Build

Feature	What to Deliver
MSME Onboarding	Business listing form. Firebase + IPFS storage.
Smart Contract	ERC-20 token deployed on Polygon Mumbai. Show Polygonscan link.
Token Issuance	Token issuance via contract. Transaction hash on Polygonscan.
Investor Dashboard	Browse listings. Buy token button (MetaMask transaction).
Portfolio View	Show investor holdings, valuation, dividend history.
Dividend Demo	MSME deposits test MATIC → auto-distribute → show results.
Risk Score	Calculate + display risk for each MSME.
Governance	Create proposal → vote → execute. On-chain.

5. TECH STACK (100% FREE, TESTNET ONLY)

Component	Tool	Notes
Smart Contracts	Solidity + Hardhat	ERC-20 standard. Free.
Blockchain	Polygon Mumbai Testnet	Near-zero fees. TESTNET ONLY.
Wallet	MetaMask + ethers.js	Free. Industry standard.

Storage	nft.storage / web3.storage	Free IPFS.
Frontend	Next.js + React + TypeScript	Free. Deploy on Vercel.
Database	Firebase Firestore	Off-chain data.

6. JUDGING CRITERIA (100 Points)

Criteria	Weight	What Judges Evaluate
Smart Contract Quality	**30%**	ERC-20 standard, logic, security, tests, Polygonscan verification
Technical Completeness	**25%**	Modules working end-to-end, blockchain connection, live Polygonscan links
Innovation & India Context	**20%**	MSME problem addressed, India-specific, regulatory awareness
UX & Accessibility	**15%**	User-friendly, mobile-responsive, error handling
Presentation & Vision	**10%**	SEBI regulatory path, business model, roadmap

KEY TAKEAWAYS

- Problem: 63 million MSMEs have NO equity access
- Solution: Blockchain micro equity tokenisation (testnet only)
- Track: Blockchain, Fintech & Emerging Technologies
- Timeline: 24 hours for MVP | Cost: ₹0 (free tools + testnet)
- Judging: Smart Contract 30%, Technical 25%, Innovation 20%, UX 15%, Presentation 10%
- Difficulty: HIGH (Solidity + Web3 + React)
- India Impact: Financial inclusion for 63 million MSMEs
- Legal: Hackathon educational only. Real product requires SEBI approval. Testnet mandatory — Mainnet = disqualification.

INNOVATHON 2026

Challenge 4 of 5

AI PoisonGuard — Adversarial Training Data Poisoning Detector for Indian Fintech & Government ML Systems

Track	Cybersecurity, AI/ML Security & Adversarial Machine Learning
Difficulty Level	HIGH (Python ML + FastAPI + React + Adversarial AI)
Team Size	3–4 members
Duration	24 hours
Hackathon Date	April 1, 2026 – April 2, 2026
Context	Hackathon & Educational Purpose Only

CRITICAL: This is a PROTOTYPE/SIMULATION for hackathon purposes ONLY. NOT a production cybersecurity tool, government-certified scanner, or commercial ML auditing product.

REGULATORY CONTEXT — India Jurisdiction

- CERT-In: Any security auditing tool deployed for government/critical infrastructure must be empanelled with CERT-In under IT Act 2000 Section 70.
- RBI AI/ML Governance Framework (2024): Requires all ML models in regulated entities to be explainable and auditable.
- DPDP Act 2023: Poisoned datasets containing PII carry additional legal exposure. Mock Data = No DPDP Burden.
- Real Compliance Path: CERT-In empanelment + RBI MRM alignment + DPDP compliance = 6-12 months minimum.

1. EXECUTIVE SUMMARY

Build an AI-powered web dashboard that detects adversarial training data poisoning attacks on ML models deployed in Indian fintech and government systems. The platform accepts an uploaded ML model and training dataset, runs multi-layer poisoning detection algorithms, and produces a visual Poison Risk Report showing affected data clusters, mislabeled samples, and anomalous weight patterns.

2. THE PROBLEM STATEMENT

2.1 Core Challenge

- India's ML-dependent critical systems — UPI (500M+ daily transactions), Aadhaar KYC (1.4B records), PMJAY fraud detection — are all trained on data sourced from third-party vendors or open government lakes.
- A data poisoning attack injects maliciously mislabeled samples into training data, causing the final model to develop hidden backdoors — appearing normal during testing but failing on specific trigger inputs.
- CERT-In reported a 300% increase in AI-targeted cyberattacks on Indian financial infrastructure in 2024-25.
- Over 97% of Indian organisations lack proper AI access controls and model validation pipelines (IBM Cost of Data Breach Report 2025).
- NO open-source, India-specific tooling exists to detect poisoning in ML models before deployment.

3. WHAT TO BUILD — Core Modules

3.1 Model & Dataset Ingestion

- Model upload: Accept .pkl (scikit-learn), .h5 (Keras), .onnx formats
- Dataset upload: Accept CSV training datasets (features + labels)
- Domain tagging: UPI Fraud / Credit Score / KYC / Govt Welfare / Custom

3.2 Poisoning Detection Engine (AI Core)

- Layer 1 — Statistical Analysis: Detect label distribution anomalies using Z-score + IQR methods
- Layer 2 — Spectral Signature Detection: Apply SVD-based spectral analysis to identify clusters of anomalous samples
- Layer 3 — Activation Clustering (AC): Train shadow model, extract activations, cluster with UMAP + HDBSCAN
- Layer 4 — Influence Function Analysis: Identify training samples with disproportionate influence on model predictions
- Layer 5 — Backdoor Trigger Scanning: Test trained model against synthetically generated trigger patterns
- IBM ART Integration: Leverage IBM Adversarial Robustness Toolbox for standardised detection algorithms

3.3 Visualisation Dashboard

- Cluster Visualisation: 2D UMAP projection of training data coloured by clean vs suspected-poisoned samples
- Label Heatmap: Distribution of labels across feature clusters — anomalous concentrations highlighted in red

- Influence Score Chart: Ranked bar chart of top-50 most anomalously influential training samples
- Suspected Poison Samples Table: Downloadable list of flagged sample IDs with risk reason

3.4 India Domain Risk Profiles

- UPI Fraud Detection Profile: Optimised thresholds for transaction feature poisoning patterns
- Credit Scoring Profile: Detects income/demographic feature manipulation
- KYC / Aadhaar Profile: Identifies face-feature or biometric poisoning patterns
- Government Welfare Profile: Flags eligibility-score manipulation typical in welfare disbursement fraud

4. MVP FOR 24-HOUR BUILD

Module	MVP (Must Have)	Stretch Goal
Model Ingestion	CSV dataset + sklearn .pkl upload	ONNX + Keras .h5 support
Statistical Analysis	Z-score + IQR label anomaly detection	Wasserstein distance + MMD
Spectral Detection	SVD-based spectral signature analysis	Full ART SpectralSignatures integration
Activation Clustering	UMAP + KMeans shadow model clustering	HDBSCAN + influence functions
Dashboard Visualisation	Scatter plot + risk score + flagged table	Animated cluster transitions + PDF export
Domain Profiles	2 profiles (UPI Fraud + Credit Score)	All 5 India-specific domain profiles

5. TECH STACK (100% FREE, DEMO ONLY)

Layer	Technology	Purpose
ML Detection Backend	Python 3.11 + scikit-learn + IBM ART	Poisoning detection algorithms
Deep Learning	PyTorch / TensorFlow (lite)	Shadow model + activation clustering
Dimensionality Reduction	UMAP-learn + HDBSCAN	Cluster visualisation
API Layer	FastAPI + Uvicorn	REST endpoints for frontend

Frontend	React 18 + Vite	Dashboard UI
Visualisation	Recharts + D3.js	Cluster plots, risk charts
Dataset	NSL-KDD / CICIDS2017 / Synthetic UPI CSV	Demo poisoning scenarios

6. JUDGING CRITERIA (100 Points)

Criteria	Weight	What Judges Evaluate
Detection Engine Quality	**30 pts**	Accuracy of poisoning detection, layers implemented, false positive rate on clean data
Technical Implementation	**25 pts**	Code quality, API design, ML pipeline robustness, error handling
Innovation & Uniqueness	**20 pts**	Novel detection approach, India-specific context, originality of solution
Dashboard UX	**15 pts**	Clarity of visualisations, usability, quality of risk report output
Presentation & Demo	**10 pts**	Live demo quality, ability to explain detection methodology, CERT-In/RBI awareness

KEY TAKEAWAYS

- Problem: India's critical ML systems (UPI, Aadhaar, PMJAY) have no accessible poisoning detection tooling
- Solution: AI-powered poisoning detection dashboard — multi-layer adversarial ML analysis (demo only)
- Track: Cybersecurity, AI/ML Security & Adversarial Machine Learning
- Timeline: 24 hours for MVP | Cost: Rs. 0 (free tools + open datasets)
- Judging: Detection Engine 30%, Technical 25%, Innovation 20%, UX 15%, Presentation 10%
- Difficulty: HIGH (Python ML + FastAPI + React + Adversarial AI knowledge required)
- India Impact: Protecting 500M+ daily UPI transactions and 1.4B Aadhaar records from silent AI sabotage

- Legal: Hackathon educational only. Real product requires CERT-In empanelment + DPDP Act + RBI MRM compliance.
- Ethics: No real system testing. No real data. Responsible disclosure norms apply.

INNOVATHON 2026

BioShield IoT — Cancellable Fingerprint Template System for Identity-Theft-Proof User Verification on IoT Devices

Track	IoT Security, Biometric Template Protection & Privacy-Preserving Authentication
Difficulty Level	HIGH (Android + BioHashing + FastAPI + React + Cryptography)
Team Size	3–4 members
Duration	24 hours
Hackathon Date	April 1, 2026 – April 2, 2026
Context	Hackathon & Educational Purpose Only

CRITICAL: This is a PROTOTYPE/SIMULATION for hackathon purposes ONLY. NOT a real biometric authentication product, UIDAI-certified system, or commercially deployable identity verification service.

REGULATORY CONTEXT — India Jurisdiction

- Aadhaar & UIDAI Regulation: Any real biometric authentication system in India using fingerprints is REGULATED by UIDAI under the Aadhaar Act 2016. This prototype does NOT integrate with UIDAI or Aadhaar infrastructure.
- DPDP Act 2023: India's Digital Personal Data Protection Act classifies biometric data as sensitive personal data. Any real deployment requires explicit consent, data minimisation, and breach notification.
- IT Act 2000 Section 43A: Negligent handling of sensitive personal data (including biometrics) carries civil liability.
- No Real Biometric Collection: This prototype does NOT collect, store, or process real human fingerprint data. All fingerprint inputs are simulated via Android Emulator or FVC2002 public dataset samples.
- Real Compliance Path: UIDAI AUA certification + DPDP Act compliance + BIS IoT security certification = 12-18 months minimum.

1. EXECUTIVE SUMMARY

Build a cancellable fingerprint template system for IoT device authentication — where fingerprint minutiae are transformed using a non-invertible, user-specific mathematical function before storage, so that even if the template database is breached, original fingerprints cannot be reconstructed, and compromised templates can be instantly cancelled and re-issued using a new transformation key.

The system simulates an IoT access control scenario (smart lock / industrial entry gate) where a user enrolls their fingerprint on a virtual IoT device (Android Emulator), the cancellable template is generated and stored, and verification happens entirely in the transformed domain — the original biometric never leaves the edge device.

2. THE PROBLEM STATEMENT

2.1 Core Challenge

- Conventional biometric systems store fingerprint templates as fixed, immutable records — once stolen, the victim's identity is permanently compromised. Unlike a password, you cannot issue a new fingerprint.
- India's smart city, industrial IoT, and access control deployments store fingerprint templates in centralised databases — creating high-value targets for attackers.
- CERT-In 2024: Biometric data breaches increased 340% YoY in India. Over 93% of IoT-based biometric deployments lack any template revocation mechanism.
- There is NO accessible, open-source cancellable fingerprint template system designed for resource-constrained IoT devices in India.

2.2 India-Specific Context

- 100 Smart Cities Mission: Thousands of IoT-enabled access control systems being deployed with biometric authentication — mostly without template protection.
- UIDAI's 1.4 billion Aadhaar fingerprint records represent the world's largest single biometric database — making India disproportionately exposed to large-scale biometric breach risk.
- Make in India IoT: India's IoT device market projected to reach $9.28 billion by 2025 (NASSCOM) — majority lack biometric template protection standards.

2.3 What is Cancellable Biometrics?

Cancellable biometrics transforms the original fingerprint feature vector using a non-invertible, user-specific function BEFORE storing it. Three key properties must be satisfied:

Property	Definition	Why It Matters
Non-Invertibility	Cannot reconstruct original fingerprint from stored template	Stolen template is useless to attacker
Revocability	Change the transformation key = completely new template from same	Compromised template can be cancelled instantly

	fingerprint	
Unlinkability	Same fingerprint with different keys produces unrelated templates	Cannot link user across multiple databases
Performance	Verification accuracy remains high in transformed domain	Security does not sacrifice usability

3. WHAT TO BUILD — Core Modules

3.1 Fingerprint Input & Feature Extraction

- Emulator Input: Use Android Studio Extended Controls to simulate fingerprint touch events on virtual IoT device
- Dataset Fallback: Use FVC2002 DB1/DB2 public fingerprint dataset for batch testing
- Minutiae Extraction: Extract fingerprint ridge endings and bifurcations using SourceAFIS library
- Feature Vector Generation: Convert minutiae points (x, y, angle, type) into a fixed-length numerical feature vector

3.2 Cancellable Template Generation Engine (Core Innovation)

- User Key Generation: Generate a unique, user-specific random transformation key (stored separately from template)
- Option A — BioHashing: Project minutiae feature vector onto a random subspace using user key, then binarise. Fast, lightweight, proven.
- Option B — Random Projection + XOR: Apply random projection matrix then XOR with key-derived binary mask. Stronger unlinkability.
- Non-Invertibility Guarantee: Transformation is many-to-one — making reconstruction mathematically infeasible
- Key Separation: User key stored in separate AES-256 encrypted key vault — never co-located with template

3.3 IoT Device Simulation (Android Emulator)

- Android Emulator Setup: Pixel 6 API 33 AVD with fingerprint sensor enabled in Extended Controls
- Enrollment Flow: User places finger on virtual sensor → minutiae extracted → cancellable template generated → encrypted template stored
- Verification Flow: User authenticates → new reading transformed with stored key → Hamming distance match → ACCEPT / REJECT
- IoT Scenario: Frame as smart door lock — LED indicator (green = access granted, red = denied) simulated on screen

3.4 Template Cancellation & Re-Enrollment

- Cancel Trigger: User reports compromise → system revokes current transformation key
- Old Template Invalidation: Previous template marked as cancelled — all future matches against it will fail

- Re-Enrollment: User re-enrolls same fingerprint → completely different cancellable template generated using new key
- Unlinkability Demo: Show that old template and new template are mathematically unrelated

3.5 Web Dashboard (Visualisation & Control Panel)

- Enrollment Panel: Trigger enrollment on emulator, show minutiae extraction result, display generated cancellable template
- Template Visualisation: Side-by-side view of original minutiae map vs transformed template
- Security Metrics: Live FAR/FRR graphs, match threshold slider, transformation strength indicator
- Attack Simulation Panel: Show what happens if attacker steals database — demonstrate template is useless without key

4. MVP FOR 24-HOUR BUILD

Module	MVP (Must Have)	Stretch Goal
Fingerprint Input	Android Emulator simulated fingerprint	FVC2002 dataset batch enrollment
Feature Extraction	SourceAFIS minutiae extraction (Java)	Custom Gabor filter preprocessing
Transformation	BioHashing (Random Projection + binarise)	Cartesian Domain Transform
Template Storage	AES-256 encrypted SQLite / JSON vault	Hardware-backed keystore simulation
Cancellation	Key rotation + template invalidation	Full audit trail + revocation certificate
Verification	Hamming distance matching in transformed domain	Fuzzy vault matching scheme
Dashboard	Enrollment + Verify + Cancel flow with template viz	FAR/FRR graph + attack simulation panel
IoT Simulation	Android Emulator virtual smart lock	Raspberry Pi physical prototype

5. TECH STACK (100% FREE, EMULATOR ONLY)

Layer	Technology	Purpose

IoT Device Simulation	Android Studio + Pixel 6 AVD (API 33)	Virtual IoT fingerprint node
Fingerprint SDK	SourceAFIS (Java / Python port)	Minutiae extraction from fingerprint image
Transformation Engine	Python 3.11 + NumPy + SciPy	BioHashing / Random Projection transform
Key Management	PyCryptodome (AES-256)	User key generation and secure vault
Mobile App	Android (Java/Kotlin) + BiometricPrompt API	Emulator fingerprint enrollment UI
Backend API	FastAPI + Uvicorn	Template storage, verification, cancellation endpoints
Frontend Dashboard	React 18 + Vite + Recharts	Visualisation and control panel

6. JUDGING CRITERIA (100 Points)

Criteria	Weight	What Judges Evaluate
Cancellable Template Implementation	**30 pts**	Correctness of non-invertible transform, revocability demonstration, unlinkability proof
Technical Implementation	**25 pts**	Code quality, Android + API integration, security architecture, key separation
Innovation & IoT Context	**20 pts**	Creativity of approach, India relevance, IoT scenario framing, attack resistance
Dashboard & Visualisation	**15 pts**	Quality of template viz, demo flow clarity, FAR/FRR display, UX
Presentation & Security Awareness	**10 pts**	Live demo quality, ability to explain non-invertibility, UIDAI/DPDP awareness

KEY TAKEAWAYS

- Problem: Once a fingerprint is stolen from a conventional IoT system, the victim's identity is permanently compromised — there is no recovery
- Solution: Cancellable fingerprint templates — non-invertible transformation means stolen templates are useless, compromised identities can be re-issued
- Platform: Android Emulator (virtual IoT smart lock) + Python backend + React dashboard
- Track: IoT Security, Biometric Template Protection & Privacy-Preserving Authentication
- Timeline: 24 hours for MVP | Cost: Rs. 0 (Android Studio + free libraries + emulator)
- Judging: Template Implementation 30%, Technical 25%, Innovation 20%, Dashboard 15%, Presentation 10%
- Difficulty: HIGH (Android + BioHashing math + FastAPI + React + cryptography)
- India Impact: Protecting 1.4B Aadhaar records + 100 Smart Cities IoT biometric deployments
- Legal: Hackathon educational only. Real product requires UIDAI AUA certification + DPDP Act compliance + BIS IoT standard.
- Ethics: NO real fingerprint collection. Android Emulator + FVC2002 dataset ONLY.